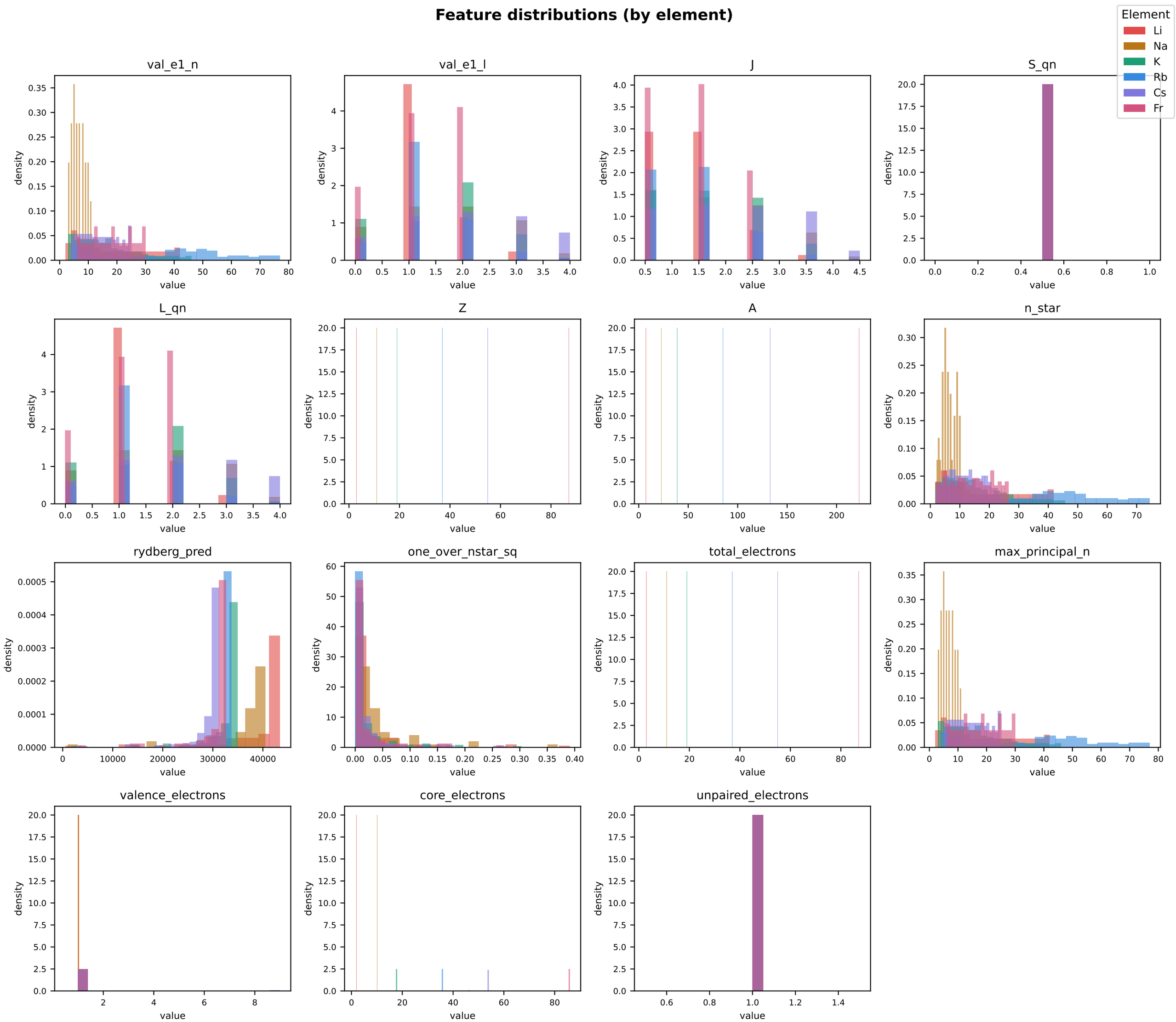
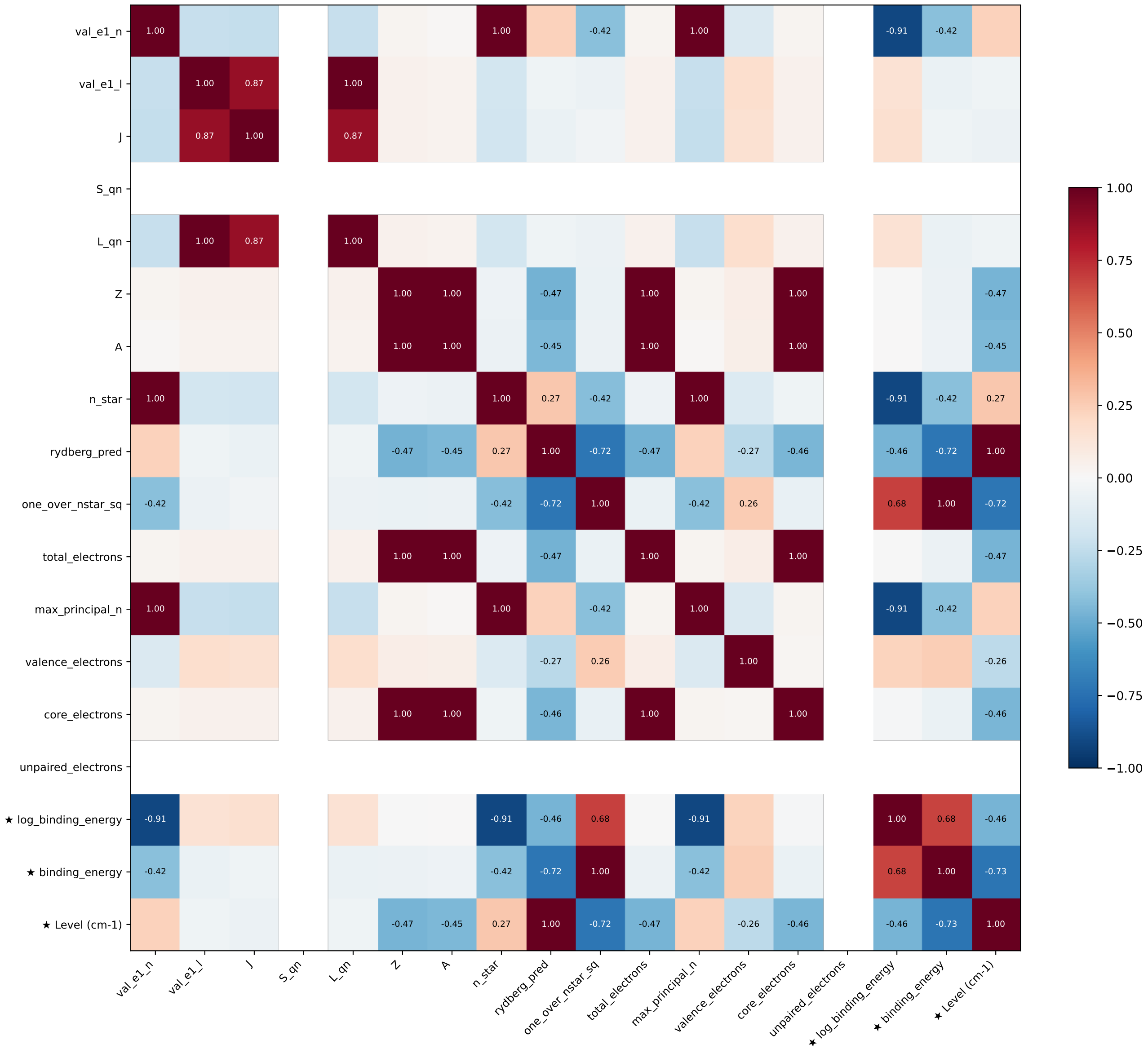
# Feature Analysis Report

Elements: Li, Na, K, Rb, Cs, Fr · n=844 total samples

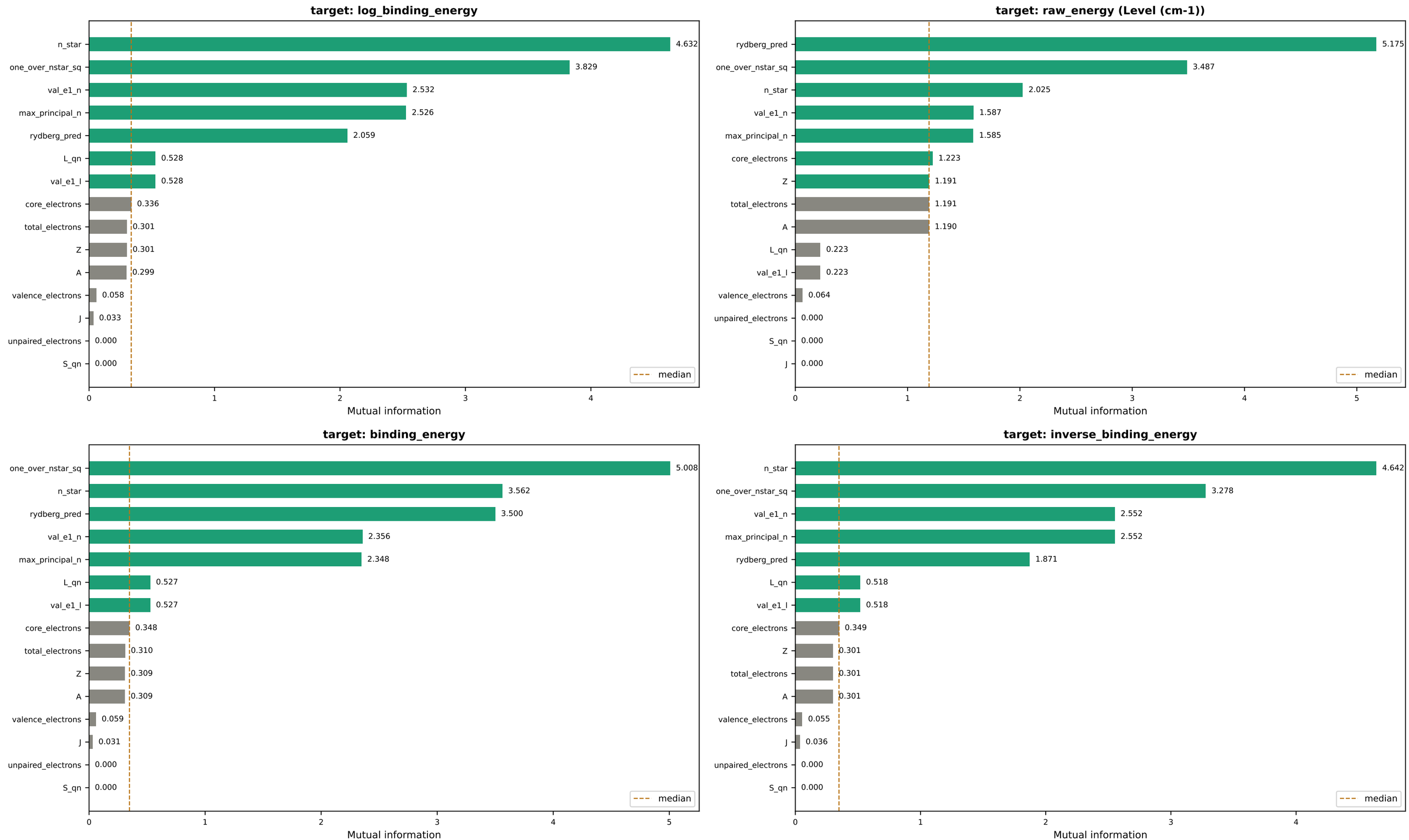
Features analysed: val_e1_n, val_e1_l, J, S_qn, L_qn, Z, A, n_star, rydberg_pred, one_over_nstar_sq, total_electrons, max_principal_n, valence_electrons, core_electrons, unpaired_electrons


Feature distributions (by element)
Element
Li
Na
K
Rb
Cs
Fr
val_e1_n
val_e1_l
J
S_qn
L_qn
Z
A
n_star
rydberg_pred
one_over_nstar_sq
total_electrons
max_principal_n
valence_electrons
core_electrons
unpaired_electrons
density
value

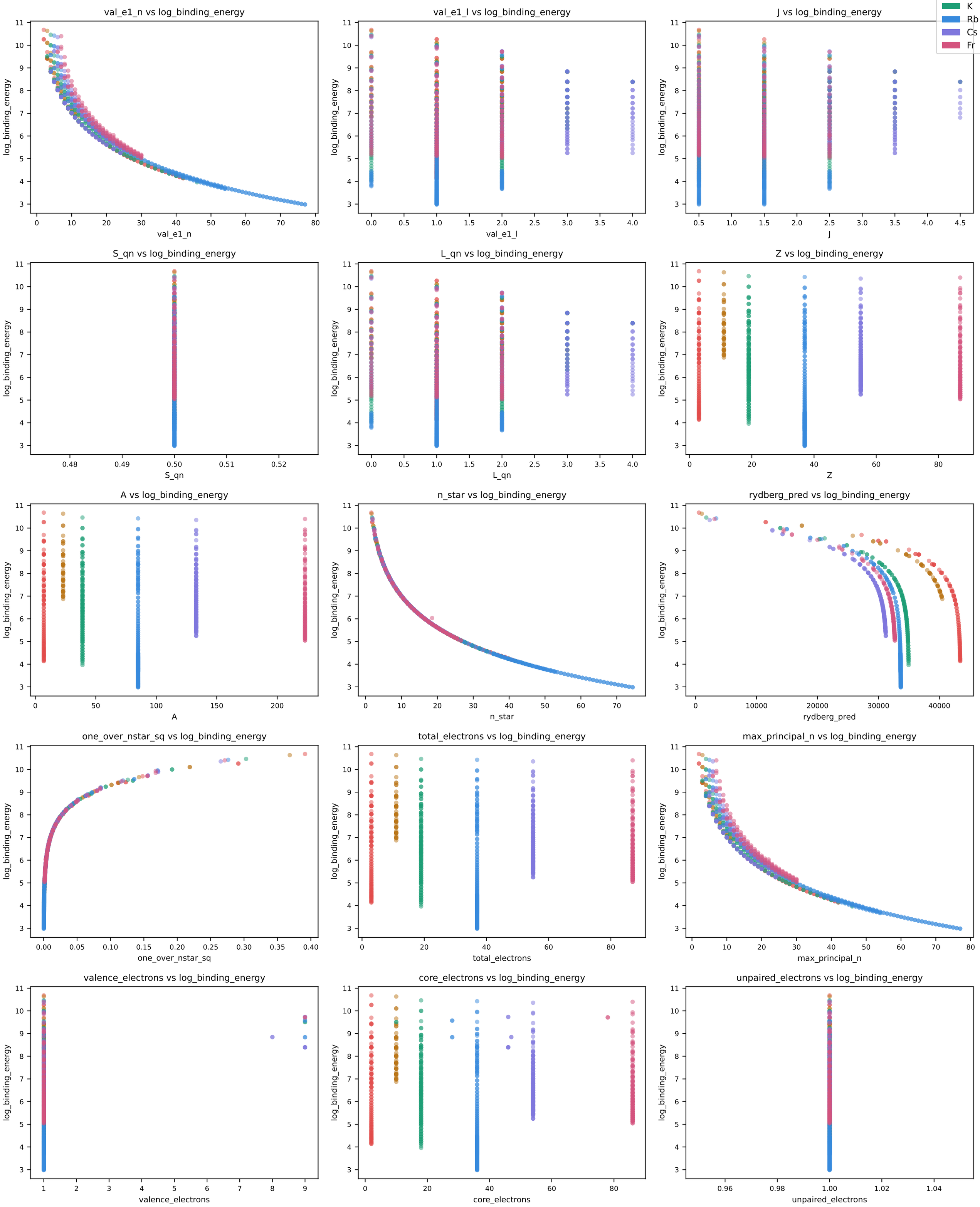
**Feature correlation matrix (★ = target variables)**
**|r|>0.8 suggests redundancy; look for L_qn≈val_e1_l, S_qn constant**



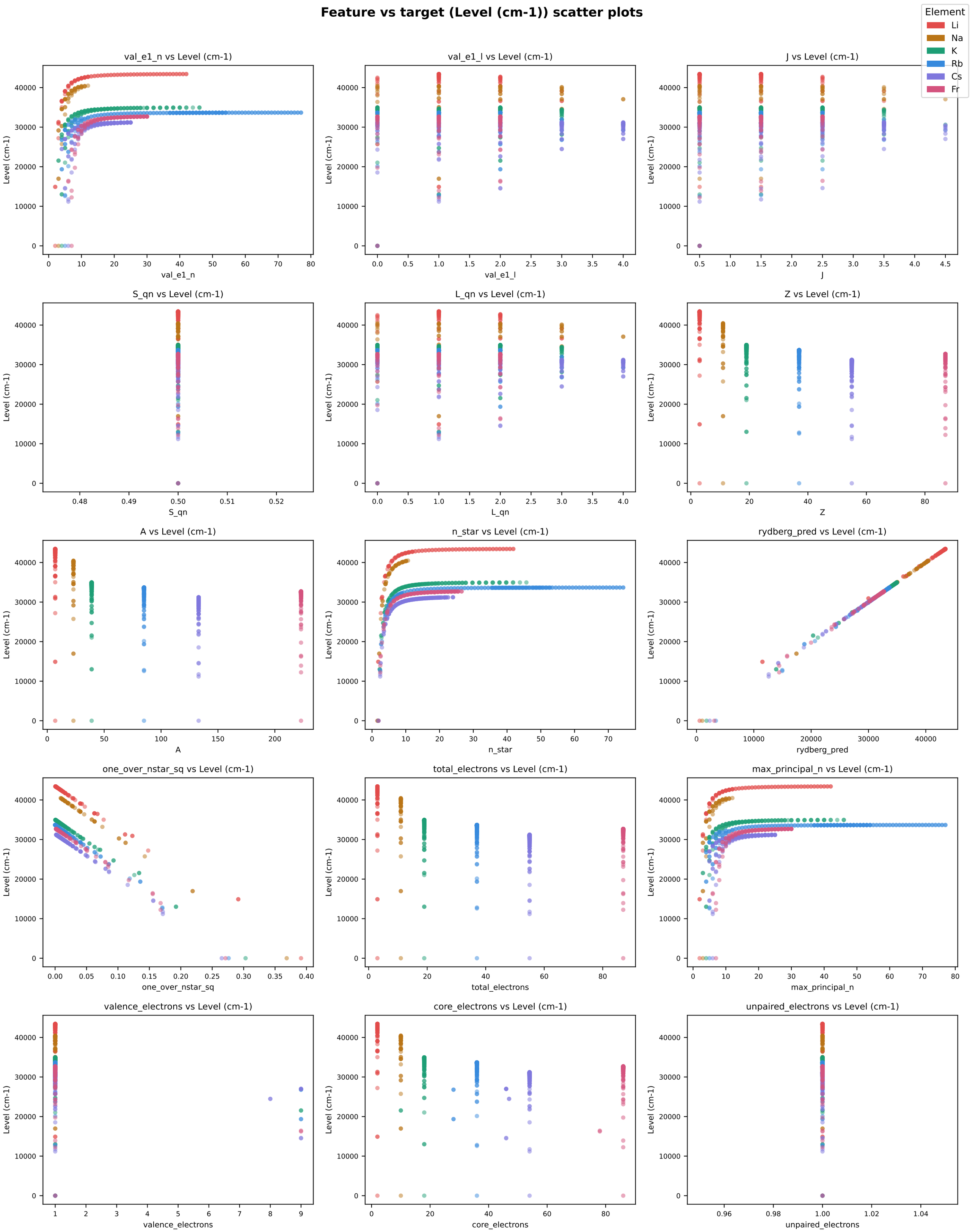
# Mutual information ranking — feature importance per target
(higher = more informative; near-zero = essentially noise for that target)



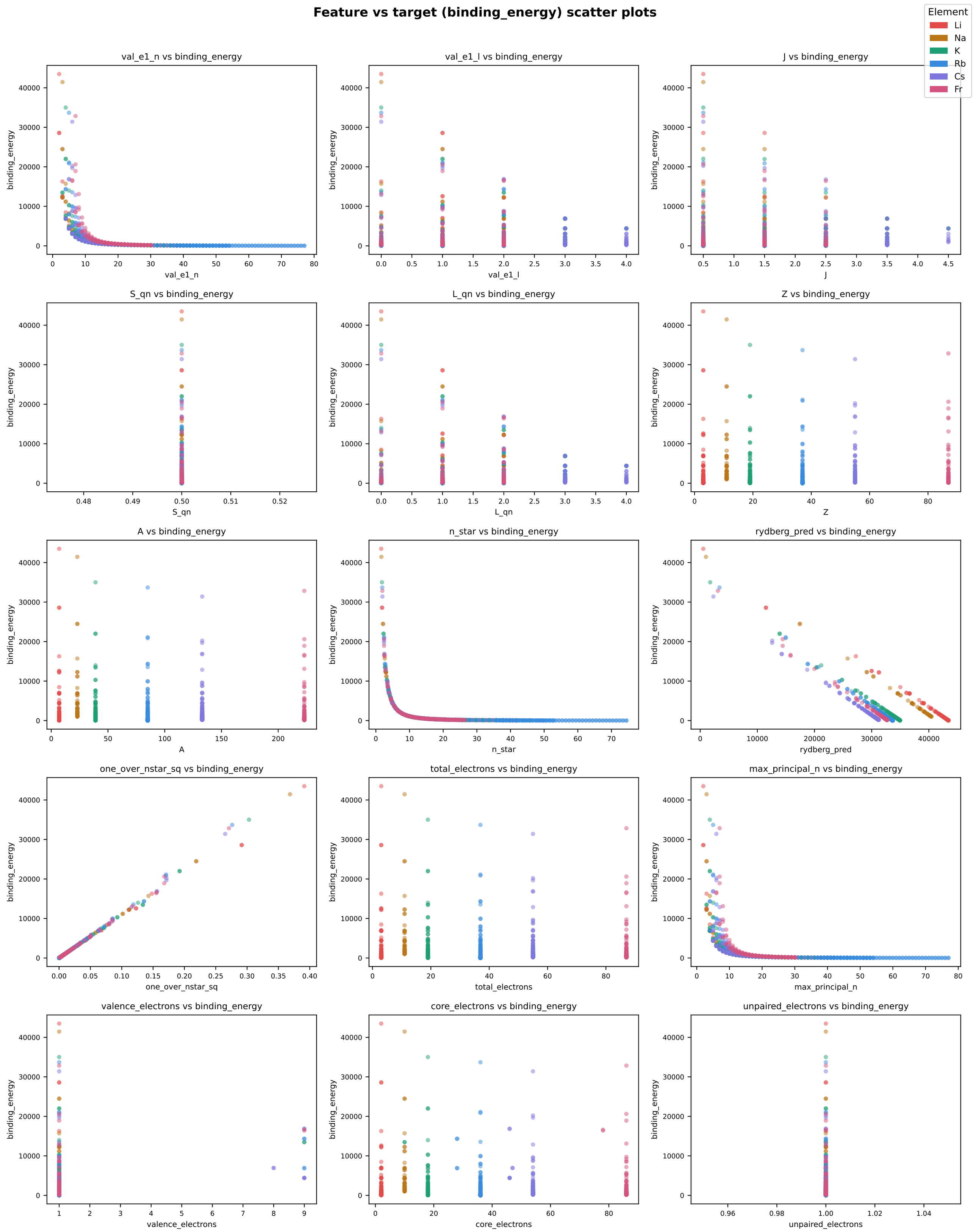
# Feature vs target (log_binding_energy) scatter plots



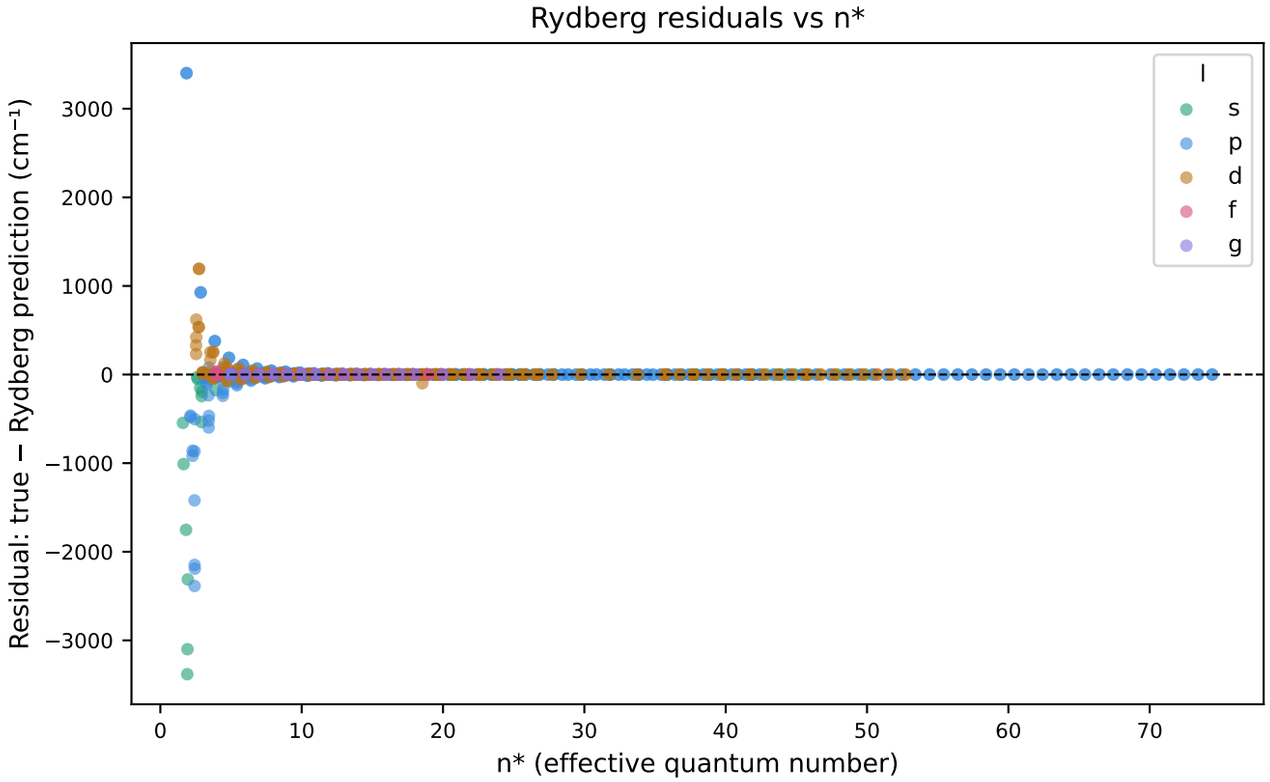
# Feature vs target (Level (cm-1)) scatter plots

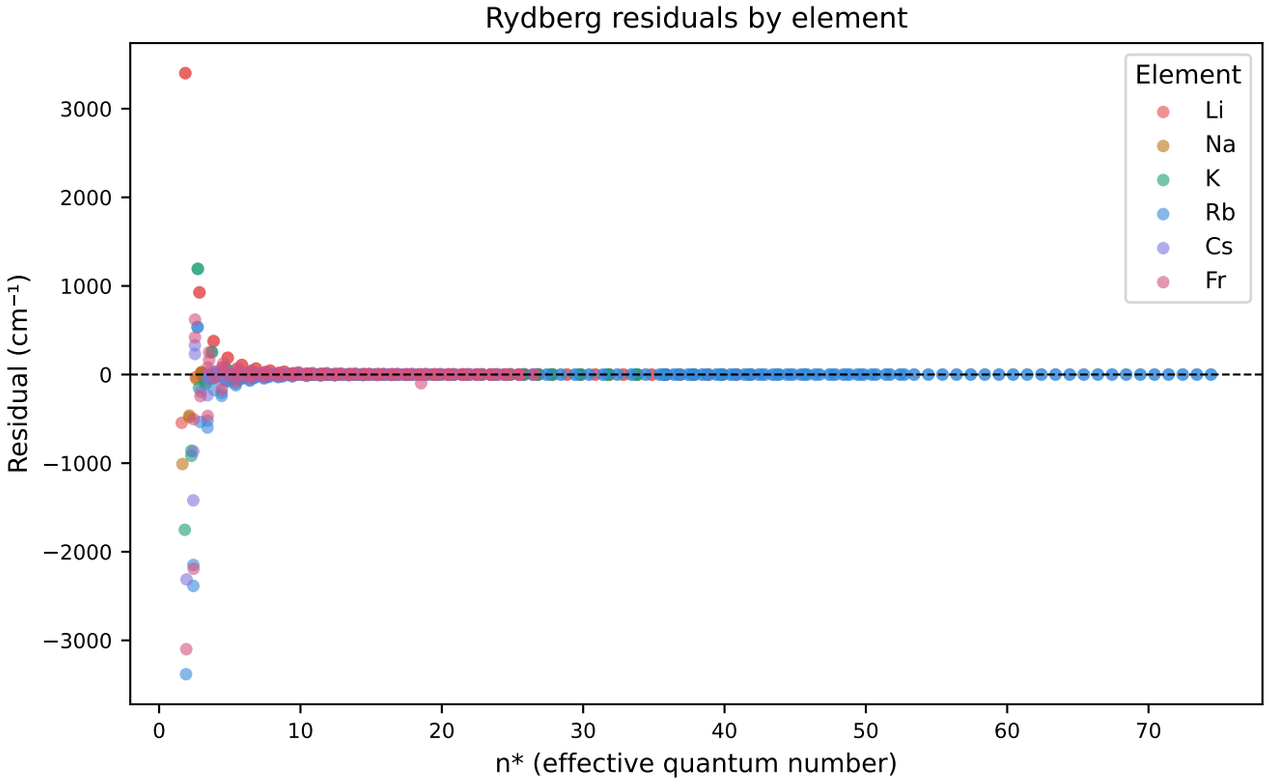


# Feature vs target (binding_energy) scatter plots

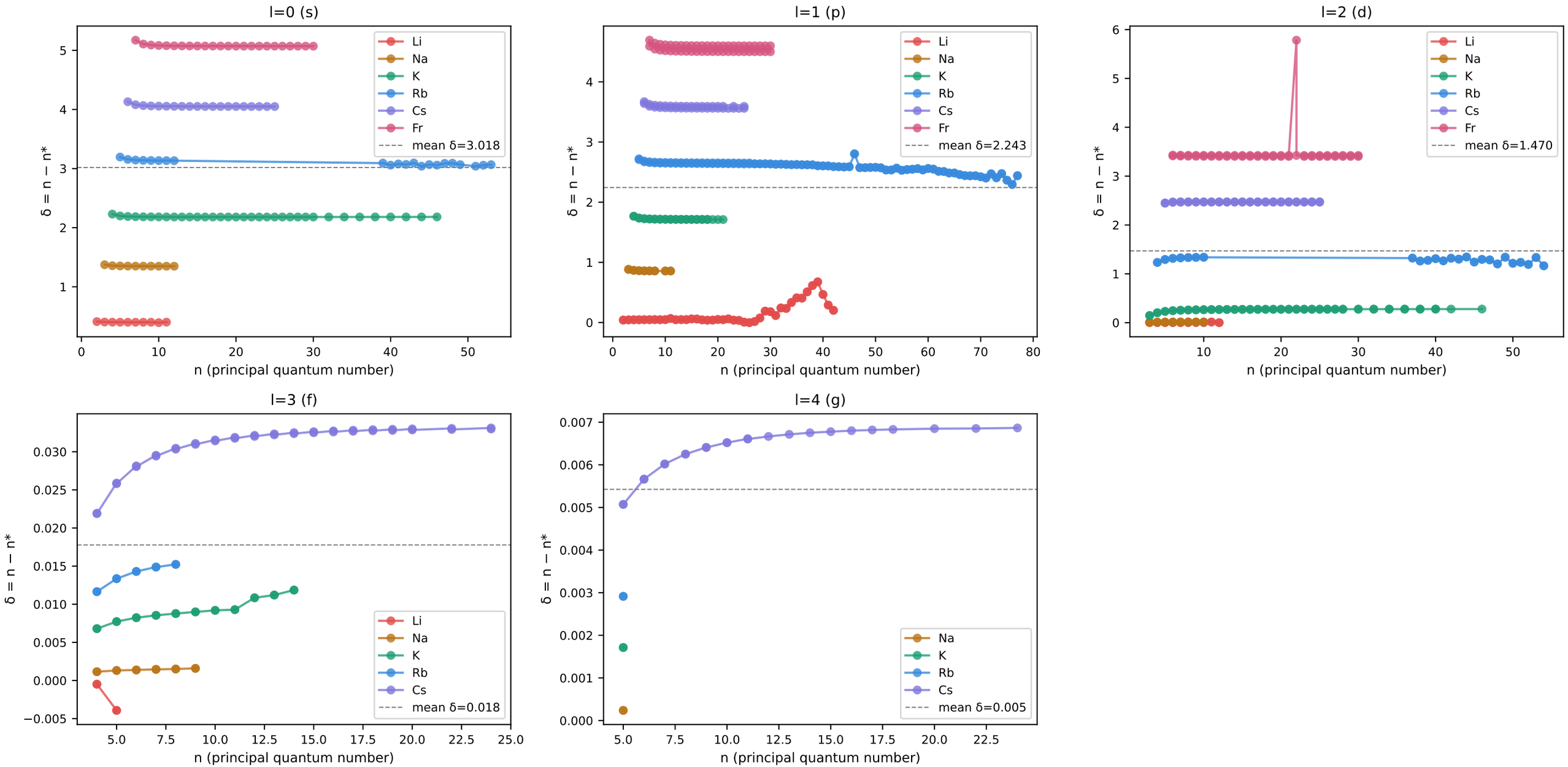


**Rydberg residuals (true level − Rydberg prediction)**
**Large residuals at low n* = extrapolation challenge; ideally residuals should be small and symmetric around 0**

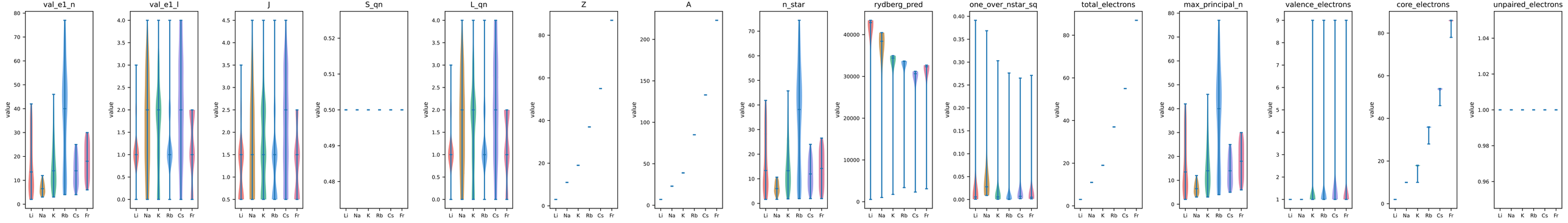
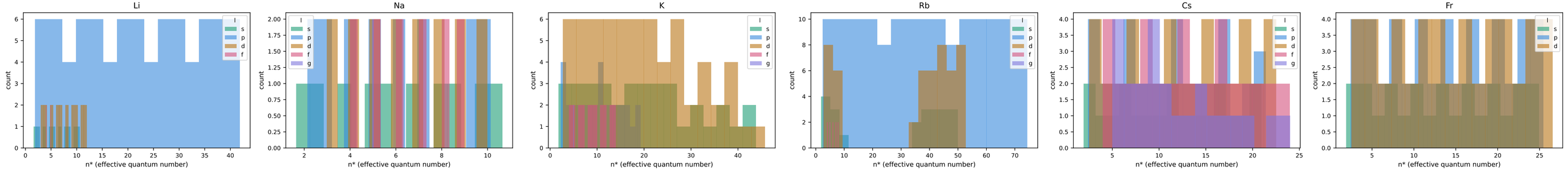


**Quantum defect δ = n − n* per angular momentum series**
**Ideally a flat horizontal line; slope indicates Rydberg-Ritz correction needed**



**Feature value ranges per element (violin plots)**
**Mismatched ranges across elements can confuse StandardScaler — look for outliers**



**n* (effective principal quantum number) distribution per element and series**
**Gaps at low n* identify states where Rydberg extrapolation is required**